Innovation Diffusion of Camp Snap: A Bass Model Analysis

DS 223 Marketing Analytics: Homework on Bass Model

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1. The innovation

Camp Snap - a screen-free digital camera, one of TIME's Best Inventions of 2025.

- TIME entry: https://time.com/collections/best-inventions-2025/7318270/camp-snap/

Camp Snap is a simple, child-friendly point-and-shoot digital camera with no screen: like a disposable film camera, you take the picture and only see it later, after downloading. One USB-C charge lasts about 500 shots. It was created by parents Brian and Melanie Waldman for children at summer camps that ban phones, and TIME highlights it as a response to the spread of screen-free policies in schools and camps [1]. It launched in 2023, costs around US$65 [2], and has sold more than one million cameras, mostly through its own online store (about 90 % of sales) plus US retailers such as Target, REI and Best Buy [3].

2. Look-alike innovation: the digital still camera

The closest historical analogue is the **digital still camera**. Functionally, Camp Snap *is* a digital camera: it captures images on a sensor and stores them on a memory card, and its core technology (CMOS sensor, digital storage, USB transfer) is what digital compact cameras brought to the mass market from the late 1990s. The difference is a design choice - it deliberately removes the screen - not a new technology.

In market terms the two are also similar: both are inexpensive-to-mid-priced consumer electronics bought for personal and family memories, sold through the same channels, and spread largely by word of mouth and visible use (people see the photos, then buy the device). Camp Snap's own growth has been driven mainly by word of mouth, including a viral moment when it was seen with Taylor

Swift [3] - the kind of imitation effect the Bass q coefficient captures. The main difference is scale: the digital camera became a near-universal product, while Camp Snap is a niche product for children, families and nostalgia buyers. So we take the **imitation coefficient q** from the look-alike, but estimate Camp Snap's own **market potential M** and calibrate its **innovation coefficient p** to its actual sales.

3. Historical data

`data/cameras_and_lenses.csv` contains annual worldwide camera shipments (thousand units, 1951–2020) by member companies of the Camera & Imaging Products Association (CIPA, Japan), as compiled by Statista [4][5]. Shipments peaked at 121 million units in 2010 and fell to 8.9 million by 2020 as smartphones replaced compact cameras.

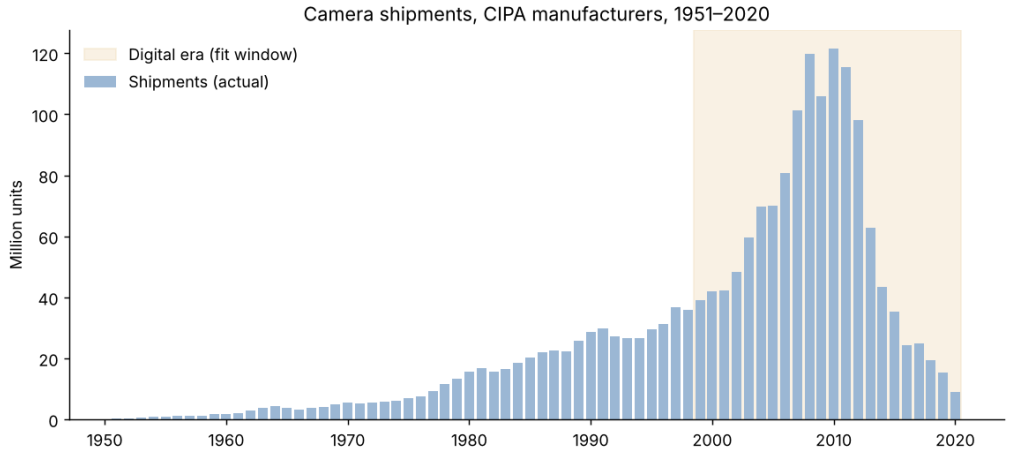


Figure 1: Camera shipments 1951–2020; shaded = digital-era fit window

The series spans two generations - film cameras (1951–late 1990s) and digital cameras. Since CIPA began reporting digital still cameras in 1999 and digital models dominated shipments from the early 2000s, the Bass model is fitted on **1999–2020**, the life cycle of the digital camera. The full 1951–2020 fit is reported for comparison.

4. Bass model parameter estimation

The Bass model states that the hazard of adoption is $\frac{f(t)}{1-F(t)} = p + qF(t)$, giving

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}, \qquad a(t) = M \cdot f(t) = M\frac{(p+q)^2}{p}\frac{e^{-(p+q)t}}{\left(1 + \frac{q}{p}e^{-(p+q)t}\right)^2}.$$

Two estimation methods from the lecture were used: (1) non-linear least squares on $a(t)$ (the Python equivalent of R's `nls()`), and (2) Bass's (1969) linear regression $a(t) = \beta_0 + \beta_1 A(t-1) + \beta_2 A(t-1)^2$ with $p = \beta_0/M$, $q = -M\beta_2$.

Series	Method	M (million units)	p	q	R^2	Peak (years after start)
Cameras 1999–2020	**NLS**	**1,374.6**	**0.0106** (se 0.0017)	**0.3096** (se 0.0203)	**0.91**	**10.5**
Cameras 1999–2020	OLS (Bass 1969)	1,367.9	0.0163	0.2951	–	9.3
Cameras 1951–2020	NLS	1,980.0	0.0003	0.1064	0.62	55.4
Smartphones 2007–2020 (benchmark)	NLS	17,449.7	0.0053	0.3659	0.99	11.4

The digital-era fit is good ($R^2 = 0.91$) and both methods agree (p 0.011–0.016, q 0.30). The values are consistent with Sultan, Farley & Lehmann's (1990) meta-analysis averages (p 0.03, q 0.38) [6]: q p, so diffusion was driven mainly by imitation. The full-history fit is much worse ($R^2 = 0.62$) because it forces two technology generations into one curve, which confirms the choice of window. The smartphone benchmark from the lecture gives similar p and q.

5. Predicting Camp Snap's diffusion

Rather than copying all three parameters, the prediction combines the analogue with Camp Snap's own data:

- **q = 0.3096** - imitation coefficient from the digital-camera fit (word-of-mouth behaviour is the most transferable part of an analogy).
- **M** - Fermi estimate for the chosen market (Section 7).

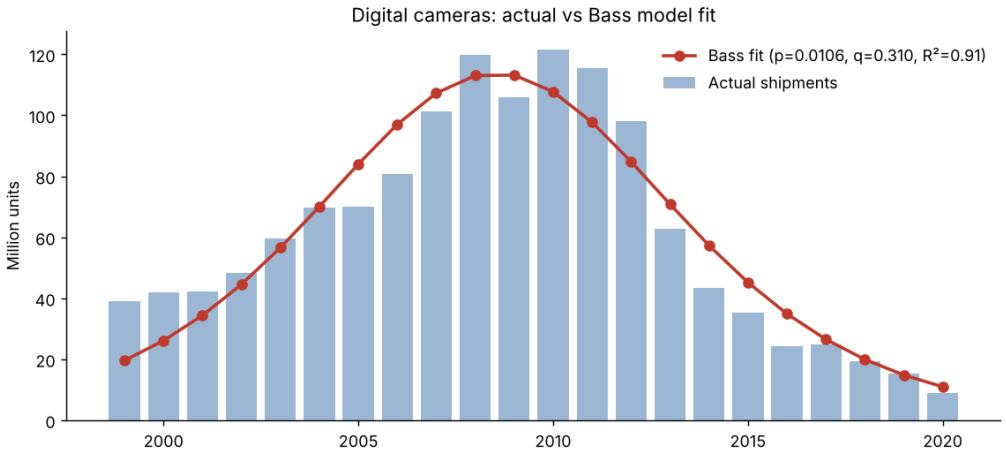


Figure 2: Digital cameras: actual vs fitted

- **p** - calibrated so the model reproduces Camp Snap's observed sales: launched 2023, **more than 1 million cameras sold** by early 2026 [3]. Setting $M \cdot F(3) = 1,000,000$ (end of 2025, $t = 3$) and solving for p gives $\mathbf{p = 0.0178}$ in the base case.

The calibrated p lies between the two analogue estimates (0.0106 NLS, 0.0163 OLS), which suggests the analogy is reasonable: Camp Snap has diffused at roughly the speed digital cameras did, slightly faster at the start.

6. Scope: United States

The analysis is **country-specific (United States)**:

- Camp Snap is a US company (founded by a US family) and its sales are concentrated in the US: about 90 % come from its own e-commerce site and the rest from US retailers (Target, REI, Best Buy, Hudson News) [3]; the company website's region selector lists only the United States [7].
- Its core use case - US summer camps and schools with screen-free policies [1] - is a strongly American market; 12 % of US children attend summer camp [8].
- A global model would require guessing shares for markets where the product is hardly sold, while US population data allow a well-referenced Fermi estimate.

The q coefficient comes from worldwide camera data; imitation behaviour for consumer electronics is broadly similar across developed markets, and p is re-calibrated on (mostly US) Camp Snap sales, so the parameters fit a US-level model.

7. Market potential and adopters by period

Fermi estimate of US market potential M (lifetime adopters):

Component	Calculation	Adopters
US population	340 million (Census, Vintage 2024) [9]	
Children under 18	340M × 21.5 % [10]	73.1M
Core: children who attend summer camp	73.1M × 12 % [8] × 50 % eventual adoption (assumption)	4.4M
Other children	(73.1M − 8.8M) × 7.5 % (assumption)	4.8M
Adults (retro/nostalgia, travel, gifts)	(340M − 73.1M) × 1 % (assumption)	2.7M
Total		**11.9M → M = 12 million**

Low and high scenarios use M = 6M and M = 20M; p is re-calibrated to the observed 1M sales in each case.

Scenario	M	p	q	Peak year	Units in peak year	Early majority from	Laggards from
Low	6M	0.0379	0.3096	2028	579,898	2024	2032
Base	**12M**	**0.0178**	**0.3096**	**2031**	**1,035,178**	**2027**	**2035**
High	20M	0.0104	0.3096	2033	1,650,467	2029	2037

Estimated adopters by year (launch 2023 = t = 1, new adopters = $M\,[F(t) - F(t-1)]$):

Year	New adopters (base)	Cumulative (base)	Low (new)	High (new)
2023	247,755	247,755	259,795	243,295
2024	327,263	575,018	331,493	325,539

Year	New adopters (base)	Cumulative (base)	Low (new)	High (new)
2025	424,982	1,000,000	408,713	431,165
2026	539,828	1,539,828	483,129	563,412
2027	666,801	2,206,629	543,449	723,387
2028	795,766	3,002,395	578,214	908,097
2029	911,571	3,913,966	579,898	1,108,276
2030	996,510	4,910,476	548,119	1,306,966
2031	**1,035,178**	**5,945,654**	489,812	1,480,346
2032	1,019,887	6,965,541	416,241	1,602,040
2033	953,727	7,919,268	338,869	1,650,467
2034	849,310	8,768,578	266,374	1,616,345
2035	724,024	9,492,602	203,676	1,506,255
2036	594,605	10,087,207	152,466	1,340,095
2037	473,529	10,560,736	112,328	1,143,986
2038	367,955	10,928,692	81,789	942,615
2039	280,497	11,209,189	59,045	754,299
2040	210,722	11,419,911	42,362	589,599

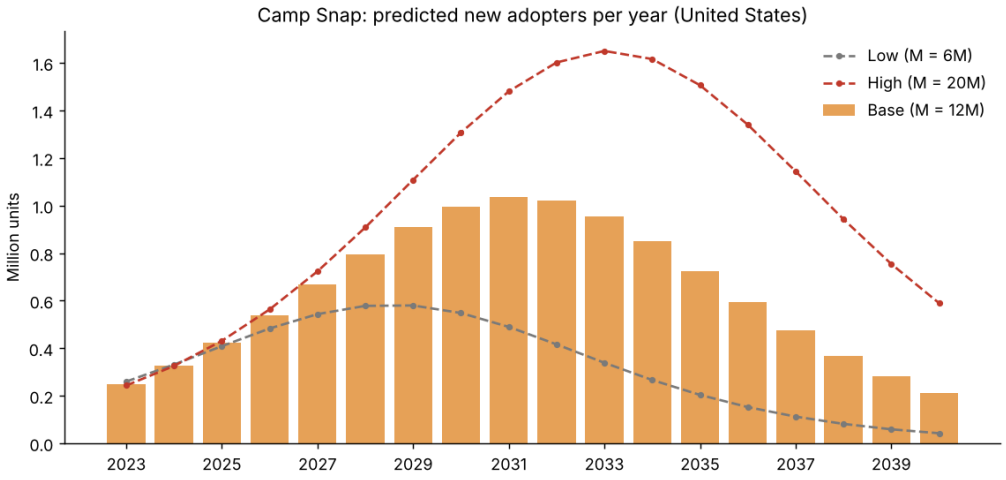


Figure 3: Predicted new adopters per year

Interpretation. In the base case, Camp Snap is still early in its life cycle: about 8 % of its US potential had adopted by the end of 2025. Yearly adoptions keep rising until a peak of about **1.0 million cameras in 2031** ($t^* = \ln(q/p)/(p+q) \approx$ 8.7 years after launch), and half of the market (6M) is reached around 2031–2032. Using the Rogers adopter categories implied by the Bass curve (boundaries at

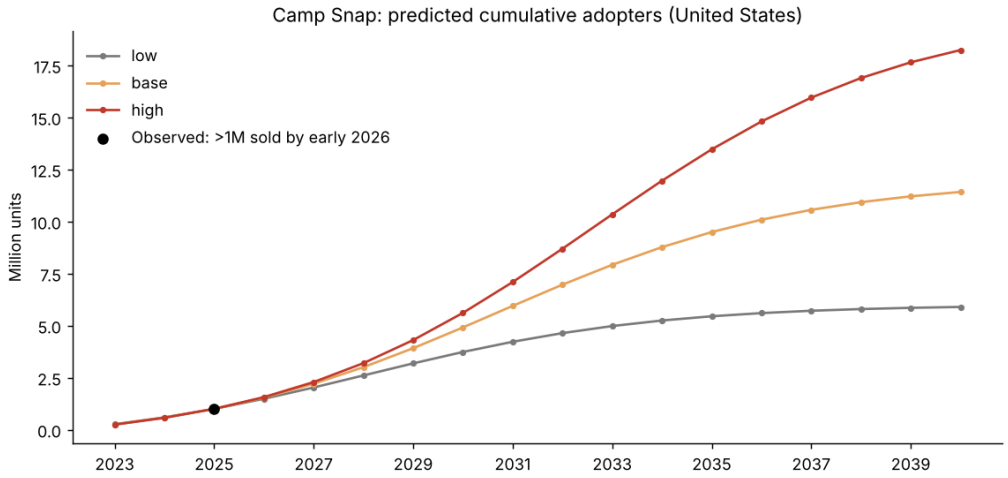


Figure 4: Predicted cumulative adopters, with the observed 1M point

$t^* \pm 1.317/(p+q)$ [11]): innovators and early adopters buy until about 2027, the early majority from 2027 to 2031, the late majority from 2031 to 2035, and laggards afterwards. If the market is smaller (low case), the peak comes sooner (2028) and lower; if it is larger (high case), growth continues until 2033 - and in that case the calibrated p (0.0104) is almost identical to the digital-camera p (0.0106).

Limitations

- q comes from an analogue; a social-media-driven gadget may spread faster than digital cameras did in the 2000s.
- M is a Fermi estimate built on assumed adoption shares; the 6–20M range is the main uncertainty, and the low/high scenarios show how the timing changes with it.
- The calibration uses one reported figure ("more than 1 million" by early 2026), treated as exactly 1M at end-2025; the true number may be higher, which would point to a larger M or faster diffusion.
- Units sold are not exactly adopters (some buyers own several cameras or replace them), a standard limitation when Bass is applied to sales data.
- Competition (Kodak Charmera, other screen-free cameras) and new Camp Snap models could shift the curve.

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